

Assignment 1

ESM 262

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Introduction & Set Up

Part 1 will assess the predictive ability of spring snowpack on summer water supply. Part 2 will assess the relationship between different types of solar radiation on net primary productivity.

```
# load packages
library(tidyverse)
library(here)
library(janitor)
library(patchwork)
library(jtools)
library(MuMIn)
```

The data `p301.csv` is provided from the USFS experimental watershed in the Southern California Sierra.

```
watershed_data <- read_csv(here("ESM 262", "p301.csv")) |> clean_names()
```

Part 1

Can spring snowpack predict the summer water supply?

Often, water supply managers use the amount of snow accumulated in the springtime to predict the supply of water throughout the summer months. This analysis aims to identify the strength of snowpack as a sole predictor of water supply later in the year. Spring snowpack is defined as the measured snowpack (in m/m²) in April of the calendar year, and summer water supply is defined as the streamflow in July of the same calendar year. Then, an OLS regression compares the relationship between the amount of snow and streamflow from 1956- 1999.

```
# spring snowpack = snowpack in April of the calendar year
# summer water supply = stream flow in July of the calendar year

# Keep data of interest
snow_stream_data <- watershed_data |>
  select(month, year, snowpack, streamflow)|>
  group_by(month, year) |>
  summarize(mean_snowpack = mean(snowpack),
            mean_streamflow = mean(streamflow))
```

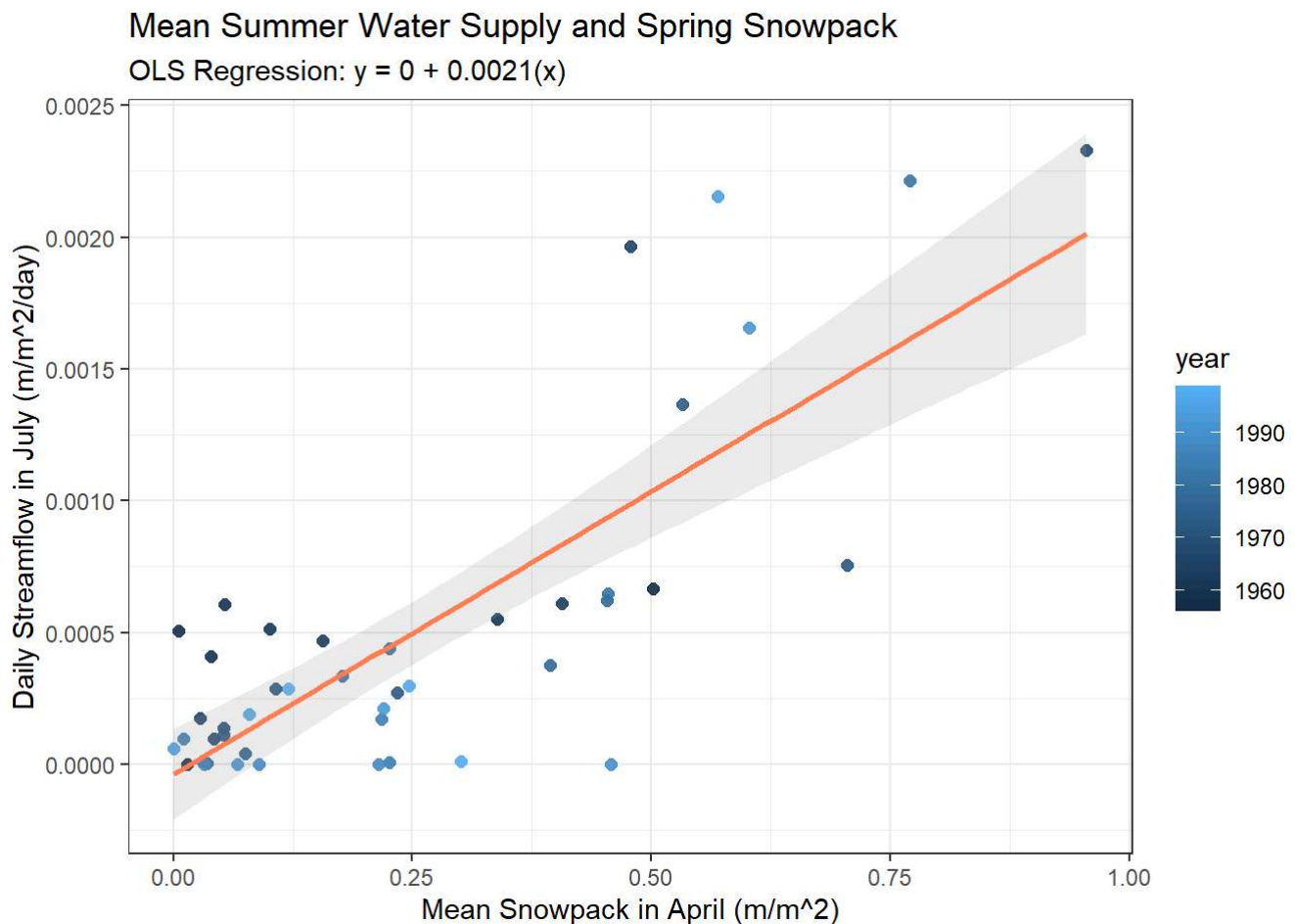


```

y = july_mean_streamflow)) +
theme_bw() +
geom_point(aes(color = year), size = 2, alpha = 0.9) +
geom_smooth(method = "lm",
            formula = y ~ x,
            color = "coral",
            size = 0.95,
            se= TRUE,
            alpha = 0.2,
            show.legend = TRUE) +
labs(title= "Mean Summer Water Supply and Spring Snowpack",
     subtitle= "OLS Regression: y = 0 + 0.0021(x)",
     x = "Mean Snowpack in April (m/m^2)",
     y = "Daily Streamflow in July (m/m^2/day)")

water_supply_plot

```



Results & Analysis

The OLS regression estimates that for every 1 m/m² increase in April snowpack, the daily July stream flow increases by 0.0021 m/m²/day. The R² value of 0.6328 suggests that approximately 63% of the change in streamflow is attributed to the change in snowpack. The standard error is relatively low as well. On paper, this model appears to represent the correlation between snowpack and streamflow well. However, the

values of measured streamflow are near zero and may be negligible in real-world water supply scenarios. However, the more research is needed to learn if this level of accuracy is appropriate for water resource management.

Part 2

Do different types of solar radiation influence net primary production (NPP)?

```
#read indata
data_direct <- read.csv("p301.csv") |>
  clean_names()|>
  select(direct_solar, npp)

data_diffuse <- read.csv("p301.csv") |>
  clean_names()|>
  select(diffuse_solar, npp)
```

Linear Regression

```
# Regression of direct solar radiation
model_direct <- lm(npp ~ direct_solar, data = data_direct)
summary(model_direct)
```

Call:

```
lm(formula = npp ~ direct_solar, data = data_direct)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0167984	-0.0022343	0.0001105	0.0022975	0.0154364

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.052e-03	5.265e-05	-19.99	<2e-16 ***
direct_solar	4.551e-07	6.001e-09	75.83	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.004263 on 16343 degrees of freedom

Multiple R-squared: 0.2603, Adjusted R-squared: 0.2602

F-statistic: 5750 on 1 and 16343 DF, p-value: < 2.2e-16

```
# Regression of diffuse solar radiation
model_diffuse <- lm(npp ~ diffuse_solar, data = data_diffuse)
summary(model_diffuse)
```

Call:

```
lm(formula = npp ~ diffuse_solar, data = data_diffuse)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0158222	-0.0023771	0.0004238	0.0023042	0.0146482

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.604e-03	7.719e-05	-59.64	<2e-16 ***
diffuse_solar	1.078e-06	1.146e-08	94.07	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

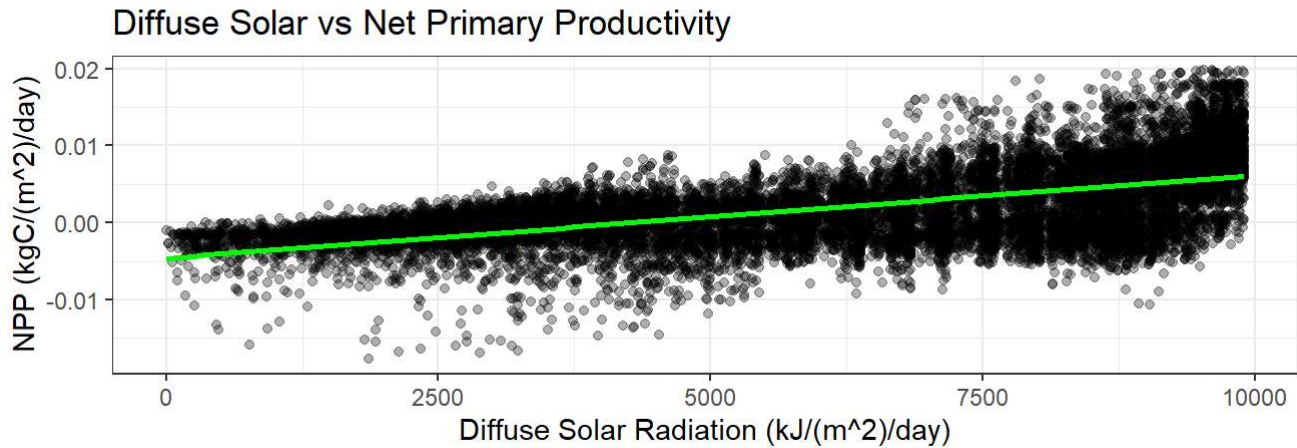
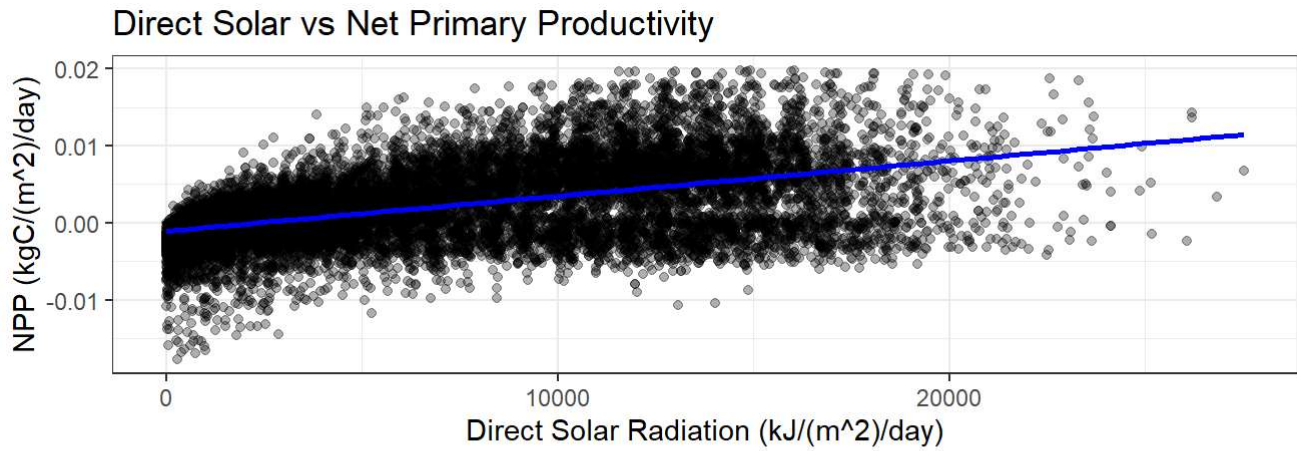
Residual standard error: 0.003992 on 16343 degrees of freedom

Multiple R-squared: 0.3513, Adjusted R-squared: 0.3512

F-statistic: 8850 on 1 and 16343 DF, p-value: < 2.2e-16

Data Visualization

```
plot_direct <- ggplot(data_direct, aes(x = direct_solar, y = npp)) +  
  theme_bw()+  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "lm",  
              color = "blue") +  
  labs(title = "Direct Solar vs Net Primary Productivity",  
        x = "Direct Solar Radiation (kJ/(m^2)/day)",  
        y = "NPP (kgC/(m^2)/day)")  
  
plot_diffuse <- ggplot(data_diffuse, aes(x = diffuse_solar, y = npp)) +  
  theme_bw()+  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "lm",  
              color = "green") +  
  labs(title = "Diffuse Solar vs Net Primary Productivity",  
        x = "Diffuse Solar Radiation (kJ/(m^2)/day)",  
        y = "NPP (kgC/(m^2)/day)")  
  
stacked <- plot_direct/plot_diffuse  
stacked
```



Results & Analysis

The goal of this analysis is to understand how different types of solar radiation influence net primary production (NPP). Specifically, comparing direct solar radiation and diffuse solar radiation to determine which better predicts variation in NPP.

Both direct and diffuse solar radiation are significantly positively related to NPP ($p < 2 \times 10^{-16}$). However, the diffuse solar radiation model explains more variation in NPP ($R^2 = 0.351$) than the model using direct solar radiation ($R^2 = 0.260$). This suggests that diffuse solar radiation may be a stronger predictor of NPP. This may be due to diffuse radiation being more evenly distributed allowing it to penetrate deeper into plant canopies, leading to more efficient photosynthesis compared to direct solar radiation.